Adarsh Kumar Jha

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WORK EXPERIENCE

AI Engineer (Production Engineering) *Jan 2026 -- Present (Intern to Full-Time, Jun 2026)*
TBO.com (Travel Boutique Online)

- Architected near-real-time **ETL/ELT** lakehouse pipelines on S3 using **Apache Spark**, **AWS Glue**, and **Kafka CDC**; enabled sub-second incremental ingestion across **4M+ daily booking events (450GB/day)**.
- Diagnosed Glue executor OOMs and partition skew, tuning shuffle partitions to 1,200 and schema caching to elevate success rate from **94% to 99.8%**, saving **$300/day**.
- Restructured **distributed data processing** by routing **ECS** reads to same-AZ **Kafka brokers**, eliminating cross-AZ network latency and saving **$5,000+/month** in AWS transfer costs.
- Engineered a schema-aware **Text-to-SQL** ontology agent validating complex **SQL** on **Amazon Athena**, cutting **BI team workload by 35%**; exposed schema tools via **MCP** to Claude clients.
- Pioneered a conversational voice bot (**AWS Nova Sonic**, **RAG**, **tool calling**) for IVR triage (<600ms p95 latency); automated classification of **650+ daily emails** via **AWS Lambda**, applying prompt caching to slash spend by **70%+**.

ML Engineer Intern *May 2025 -- Oct 2025*
IIIT-Delhi -- Research to Commercialization (R2C) | Advisor: Dr. Anuj Grover

- Built a commercialization matchmaking platform with **FastAPI REST APIs** and **multi-agent AI** document parsing, advancing project toward **INR 40 Lakhs ($50K USD) NRDC funding**.
- Calibrated document intelligence models in **PyTorch** with **LoRA** and **PEFT**, reaching **97% extraction accuracy**, **75% parameter compression**, and **2x faster GPU inference throughput**.
- Engineered an asynchronous document queue via **Redis** and **Celery**, batch-parsing **500+ PDFs daily** with zero task failures and resilient automated retries across distributed worker pools.

Independent Research Project -- Ontology Path Extraction *Aug 2024 -- Dec 2024*
IIIT-Delhi x Oracle | Advisors: Prof. Raghava Mutharaju, Suman Roy (Oracle)

- Fine-tuned **R-BERT** in **PyTorch** with span-based **embeddings**, surpassing LLaMA-3 and GPT-3 baselines by **6 F1 points**; indexed **25,000+ biomedical entities** via FAISS, reducing false positives by **20%**.
- Benchmarked relation extraction against BioBERT, reaching **0.84 micro-F1** across 10 classes on NCBI datasets.

EDUCATION

Indraprastha Institute of Information Technology Delhi (IIIT-Delhi) *2022 -- 2026*
B.Tech, Computer Science and Artificial Intelligence *CGPA: 8.2 / 10*
Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Distributed Systems, Operating Systems, DBMS, DSA

PROJECTS

VICAS Lab Portal & Resume Knowledge Graph [Code] *Aug 2025 -- Apr 2026*
React, FastAPI, Docker, AWS EC2, Neo4j Knowledge Graphs, FAISS, GCN

- Shipped a full-stack portal on **AWS EC2** across 6 microservices with OAuth2 and CI/CD, onboarding **160+ registered students** with sub-120ms endpoint latency.
- Formulated a **Gemini RAG** pipeline with **FAISS** vector retrieval and **Neo4j** (25K nodes, 140K edges); trained a **GCN** for skill prediction scoring **91.4% precision**.

ML Serving Diagnostics & Observability Platform [Code] *Jan 2025 -- Apr 2025*
Python, FastAPI, Docker, Kubernetes, TensorFlow, PyTorch, Google Cloud Storage

- Constructed an automated diagnostics engine for **Kubernetes** ML workloads, isolating root causes of OOM faults and latency spikes; slashed mean time to resolution by **40%** across 5 production services.
- Implemented a **FastAPI** telemetry dashboard streaming node health to **Google Cloud Storage**, cataloging proactive remediation runbooks that prevented service downtime.

Watch-Tower: Edge-AI Video Anomaly Detection [Code] *Aug 2024 -- Dec 2024*
Python, PyTorch, TensorRT, FastAPI, Docker, MQTT, OpenCV

- Engineered a real-time edge anomaly detection pipeline using **PyTorch** and **TensorRT**, streaming inference over **4 concurrent 1080p RTSP feeds at 30 FPS** via **MQTT** with **sub-45ms p95 latency**.
- Containerized inference microservices with **Docker** and **FastAPI**, achieving **91.4% ROC-AUC** on benchmark video datasets.

TECHNICAL SKILLS

Languages: Python, SQL (Complex Queries, Optimization, Window Functions), Java, C, C++, JavaScript
Data Systems: Apache Spark, Apache Kafka, Debezium (CDC), AWS Glue, Apache Hudi, Parquet, Amazon Athena, ETL/ELT, Data Lakehouse, DynamoDB, Neo4j, Distributed Stream Processing
AI / Machine Learning: PyTorch, LLMs, RAG, AI Agents, Tool Calling, MCP, Text-to-SQL, FAISS Vector Search, Embeddings, GCN, LoRA, PEFT, XGBoost, LightGBM, CatBoost
Cloud & DevOps: AWS (Athena, Glue, S3, Lambda, ECS, DynamoDB, CloudWatch, EC2), GCP (BigQuery), FastAPI, Docker, Kubernetes, Linux, Git, CI/CD

ACHIEVEMENTS & LEADERSHIP

Ranked AIR 252 in GATE 2026 (Computer Science & Information Technology) *2026*
JEE Mains 2022: All India Rank **4778 (99.5 percentile)** among 872,970 candidates *2022*
Indian National Physics Olympiad (INPhO): Top 5% nationally with High Distinction *2021*
Codeforces (Specialist - Max Rating 1450+), LeetCode 300+ problems solved, GFG *2023 -- 2026*
Operations Head, Odyssey IIIT-D -- Led 30-member logistics council for cultural festival hosting 30,000+ attendees *2025*